Sunina: The Language of Truth

A Universal Constructed Language

Literal Translation: Ordered-mind speech (*su* = good/order, *ni* = mind/focus, *na* = speak)
Core Philosophy: A universal, mathematically consistent language engineered for structural clarity, profound emotional expression, and logical compositionality. Every word decomposes into familiar roots that hint at its meaning — making vocabulary **memorable by design**, not arbitrary.

Contents

1 Phonology (The Universal Sounds) 2
1.1 Prosody . . . 2
1.2 Morphophonology . . . 3

2 The Periodic Table of Meaning (24 Roots) 3

3 Grammar & Syntax 3
3.1 Word Order: SOV (Subject – Object – Verb) . . . 3
3.2 Word Building (Agglutination) . . . 3
3.3 Pronouns . . . 4
3.4 Plurals & Possession . . . 4
3.5 Grammatical Particles . . . 5
3.6 Question Words . . . 5
3.7 Demonstratives . . . 6
3.8 Tense (Verb Suffixes) . . . 6
3.9 Aspect (Optional Verb Suffixes) . . . 6
3.10 Affect Particles (Speaker Stance) . . . 7
3.11 Adjectives & Adverbs (Free Modifiers) . . . 7
3.12 Comparatives . . . 8
3.13 Conditionals & Modality . . . 8
3.14 Negation . . . 8
3.15 Spatial Postpositions . . . 9
3.16 Imperative Mood . . . 9
3.17 Reflexive & Reciprocal . . . 9
3.18 Derivational Morphology . . . 9
3.19 Productive Compound Templates . . . 10
3.20 Disambiguation Conventions . . . 11

4 Core Vocabulary 13

4.1 People & Relations . . . 13
4.2 Nature & Environment . . . 14
4.3 Body & Senses . . . 15
4.4 Animals . . . 15
4.5 Food & Drink . . . 16
4.6 Core Verbs . . . 16
4.7 Actions & Objects . . . 18
4.8 Objects & Artefacts . . . 19
4.9 Time & Place . . . 20
4.10 Spatial & Direction . . . 20
4.11 Quantity & Determiners . . . 20
4.12 Qualities & States . . . 22
4.13 Emotions & Discourse . . . 23
4.14 Abstract & Mental Concepts . . . 23
4.15 Society, Colours & Health . . . 23
4.16 Proper Nouns & Loanwords . . . 24

5 Base-10 Mathematics **25**
5.1 The Number Grid (0–9) . . . 25
5.2 Place Values & Operators . . . 25

6 The Featural Writing System **25**
6.1 Base Shapes (Consonants) . . . 25
6.2 Vowel Modifiers . . . 26
6.3 Complete Glyph Table . . . 26
6.4 Orientation . . . 26
6.5 Punctuation . . . 26

7 Story Translation: "The Meeting" **27**

1 Phonology (The Universal Sounds)

Sunina uses only 11 sounds, chosen because they exist in nearly every human language. It strictly follows a **CV (Consonant-Vowel)** structure.

- **8 Consonants:** p /p/, t /t/, k /k/, m /m/, n /n/, s /s/, w /w/, y /j/
- **3 Vowels:** a /a/ (*father*), i /i/ (*machine*), u /u/ (*flute*)

1.1 Prosody

- **Rhythm:** Syllable-timed — every syllable receives equal duration (like Spanish or Hindi).
- **Stress:** Primary stress falls on the **first syllable** of the word, with secondary stress on the first syllable of subsequent roots within the compound. In *supi*, stress is on **su**. In *masupi*, primary stress is on **ma** and secondary stress on **su**. This maintains morpheme clarity without being ambiguous.

- **Intonation:** Statements fall in pitch at the end. Questions (containing *pa*) rise on the final syllable.

1.2 Morphophonology

No sound changes occur at morpheme boundaries. Each root retains its canonical CV form in all positions: *mu-ti-mu* keeps all three syllables distinct. This guarantees that roots are always recoverable by listeners, at the cost of longer words.

2 The Periodic Table of Meaning (24 Roots)

Every concept is built by combining these 24 prime syllables. The consonant defines the semantic domain; the vowel modulates it.

Consonant	-a (Open/Base)	-i (High/Internal)	-u (Round/Forward)
P (Life/Unknown)	**pa:** Question	**pi:** Life / Person	**pu:** The Unknown
T (Time/Target)	**ta:** Present / Time	**ti:** Past / Behind	**tu:** You / Target
K (Action/System)	**ka:** Do / Make / Equal	**ki:** Bond / Rule / System	**ku:** Force / Harshness
M (Self/Motion)	**ma:** Opposite / No	**mi:** I / Me / Self	**mu:** Move / Go / Change
N (Thought/Out)	**na:** Speak / Word	**ni:** Mind / Think	**nu:** Future / Forward
S (State/Reality)	**sa:** Thing / Reality	**si:** Feel / Sense	**su:** Good / Positive
W (Experience/Magnitude)	**wa:** Intense / Big	**wi:** Weight / Sorrow	**wu:** Consume / Take in
Y (Environment/Spirit)	**ya:** Joy / Harmony	**yi:** Space / Light	**yu:** Fluid / Nature

3 Grammar & Syntax

Sunina uses strict SOV word order, agglutination (stacking roots), and a minimal set of particles derived from the root system.

3.1 Word Order: SOV (Subject – Object – Verb)

- *English:* I see the friend. → *Sunina:* **mi supi yiwu.** (I friend light-consume.)
  Script: [⊖′] [Z_][◯′] [⊔′][⌣ _]

3.2 Word Building (Agglutination)

Stack roots to build the literal essence of a concept. Lexical compounds are **head-final**: the last root is the semantic head (what the word *is*), and preceding roots modify it. This mirrors SOV typology, where the core element comes last.

- **supi** = Friend (*su* [good] + *pi* [**person**] → "a good person")
- **masupi** = Enemy (*ma* [opposite] + *supi* [**friend**])
- **yusa** = Water (*yu* [fluid] + *sa* [**thing**] → "a fluid thing")
- **nisa** = Book (*ni* [mind] + *sa* [**thing**])
- **yiwu** = See (*yi* [light] + *wu* [**take in**] → "take in light")
- **siwu** = Hear (*si* [sense] + *wu* [**take in**] → "take in sound/sense")

Compound-Order Rule: In any lexical compound of n roots, the rightmost root is the head; all roots to its left narrow its meaning, outermost modifier first. E.g., *wa-ku-pi-mu* = intense > force > life > move (*pimu* = creature) = "fierce creature." (Closed-class functional series — such as question words *pa-X*, affect particles *si-X*, and directional-temporal vectors *ta-X*/*yi-X* — place their class marker first for immediate parsing.)

Shortening Rule: The official short form of a compound is always its **first two syllables** (the first stressed foot), making short forms predictable. The Sunina Academy approves short forms for any standardized compound of **3 or more syllables**; until approved, the full form is mandatory. E.g., *wakusa* (weapon) → *waku*.

Collision Avoidance: A short form must never duplicate any standardized lexical item, root, grammatical particle, or previously approved short form. If the first two syllables collide, the short form extends to the first three syllables instead (e.g., *wakupimu* [fierce creature] → *wakupi*, since *waku* is already taken by *wakusa*). If that still collides, or if the three-syllable form is no shorter than the full word, the word has no short form.

Semantic Narrowing: Compound words provide a **semantic neighbourhood**, not an exact definition. *supi* (good + person) locates the concept in the zone of "positive human relationship" — its precise meaning (*friend*) is established by the standardized lexicon. This is by design: decomposition makes vocabulary **memorable**, not self-explanatory. Much like English *understand* is not literally "standing under," Sunina compounds are mnemonic scaffolds that aid learning and recall.

Fused vs. Free Distinction: A fused compound (written as one word) carries its **standardized meaning**. The same roots written as separate words retain their literal, compositional sense. This lets speakers distinguish specialized intent from descriptive intent:

- **supi** (one word) = friend (standardized)
- **su pi** (two words) = a good person (literal)
- **masu supi** = a bad friend (modifier + standardized compound)

Stress and writing reinforce this: fused compounds receive primary stress on the first syllable as a single unit, while free modifier phrases stress each word independently.

3.3 Pronouns

- **mi** [⊖′] = I / me **tu** [△_] = you **pi** [◯′] = he / she / they **sa** [$Z\cdot$] = it

Note: *pi* and *sa* naturally double as third-person pronouns from their root meanings (person, thing). Sunina makes no gender distinction.

3.4 Plurals & Possession

- **Plural suffix *-wa*** (big → many): *pi* (person) → *piwa* (people). *nisa* → *nisawa* (books).
- **Possessive suffix *-ki*** (bond) on the possessor: *miki supi* = my friend. *tuki nisa* = your book.
  Script (my friend): [⊖′][□′] [Z_][◯′]

Note: Possessive *-ki* fuses with the possessor as one word (*miki*, stress on **mi**). The conjunction *ki* ("and") stands alone as a separate word with its own stress.

3.5 Grammatical Particles

Existing roots repurposed in grammatical positions. Role is determined by sentence context.

Function	Particle	Root Logic
And	**ki**	Bond
But / However	**maki**	Opposite-bond
Or	**paki**	Question-bond
Because	**kiti**	Bond-past (the reason behind)
That / Which / Who	**ni**	Mind-link
When / While	**taki**	Time-bond (the linked moment)
If (conditional)	**panu**	Question-forward

Examples:
mi ki tu supi ka. (I and you friends are.)
Script: [⊖′] [□′] [△_] [*Z*_][◯′] [□·]
pi ni mu-ti supi ka. (Person who came friend is.)
Script: [◯′] [$\overline{\triangle}'$] [⊖_][△′] [*Z*_][◯′] [□·]

3.6 Question Words

All wh-questions begin with **pa** (question root), combined with a domain root.

English	Sunina	Logic
What?	**pasa**	question + thing
Who?	**papi**	question + person
Where?	**payi**	question + space
When?	**pata**	question + time
Why?	**pani**	question + mind
How?	**pamu**	question + change

Wh-example: tu payi mu? (You where go? = Where are you going?)
Script: [△_] [◯·][⊔′] [⊖_] (◯·)

Yes/No (Polar) Questions: Place the particle **pa** at the *end* of a declarative sentence. Rising intonation on the final syllable reinforces the question.

- *tu mu-ti pa?* (You went? = Did you go?)
  Script: [△_] [⊖_][△′] [◯·] (◯·)
- *tasa yusa ka pa?* (This water is? = Is this water?)

Note: Wh-words appear *within* the clause (in the position of the unknown element), while polar *pa* appears strictly clause-finally. In script, (◯·) serves as the silent question mark at the end of any interrogative sentence.

3.7 Demonstratives

ta (present → proximity) and **ti** (past/behind → distance) combine with domain roots.

English	Sunina	Logic
This (thing)	**tasa**	present + thing
That (thing)	**tisa**	behind + thing
Here	**tayi**	present + space
There	**tiyi**	behind + space

Example: mi tasa wu. (I this-thing consume. = I eat this.)
Script: [⊖′] [△·][Z·] [◡ _]

3.8 Tense (Verb Suffixes)

- **Present (unmarked):** *mi sa ka.* (I make the thing.)
  Script: [⊖′] [Z·] [□·]
- **Past *-ti*:** *mi sa ka-ti.* (I made the thing.)
  Script: [⊖′] [Z·] [□·][△′]
- **Future *-nu*:** *mi sa ka-nu.* (I will make the thing.)
  Script: [⊖′] [Z·] [□·][$\overline{\triangle}$_]

Passive: Attach the suffix **-maka** (opposite-do) to the verb stem, before tense. The full verb template is **Verb(-Voice)-Tense(-Aspect)**. *sa ka-maka-ti.* = The thing was made (by someone). Binding voice to the verb prevents confusion with the free negation particle *ma*.
Copula: ka functions as "is/am/are" when linking a subject to a **noun** (classification): *pi supi ka.* = The person is a friend. *Script:* [◯′] [Z_][◯′] [□·]
Stative: The particle **sa** (reality) placed clause-finally after a modifier marks the subject as being *in a state*, distinguishing essential quality from active performance while avoiding collision with the pre-verbal object pronoun *sa* ("it"):
pi su sa. = The person is good (inherent quality). *pi su ka.* = The person does good (acts kindly).

3.9 Aspect (Optional Verb Suffixes)

Aspect describes *how* an action unfolds over time. It is **optional** — omit it when the distinction is unnecessary, preserving efficiency. When used, aspect stacks after tense (and optional voice): **Verb(-Voice)-Tense-Aspect**.

Aspect	Suffix	Root Logic	Meaning
Completed	**-ka**	do/make = done	Action is finished
Ongoing	**-mu**	move = in motion	Action is in progress
Habitual	**-ta**	time = recurring	Action repeats regularly

Examples:
mi mu-ti-ka. (I went [and arrived].) *Script:* [⊖′] [⊖_][△′][□·]

mi mu-ti-mu. (I was going [still on the way].) *Script:* [⊖′] [⊖_][△′][⊖_]
mi mu-ti-ta. (I used to go [regularly].) *Script:* [⊖′] [⊖_][△′][△·]
pi na-ti-mu taki yumu ka-ti. (The person was speaking when rain came.)
Script: [◯′] [$\overline{△}$·][△′][⊖_] [△·][□′] [⊔_][⊖_] [□·][△′]

Note: Aspect is optional. *mi mu-ti.* (I went) remains valid when the completion/duration distinction is irrelevant. Use aspect only when it adds meaning — this keeps speech efficient.

3.10 Affect Particles (Speaker Stance)

Sunina expresses *what* the speaker feels through modifier roots (§3.11). To mark *how* something is said — the speaker's stance or tone — a small set of **clause-final affect particles** is available. All are *si*-initial compounds (*si* = feel), making them instantly recognizable as tone markers. They appear *after* the complete verb phrase (including any tense/aspect suffixes).

Stance	Particle	Logic	Effect
Tenderness	**siyu**	feel + nature	Said with warmth / gentleness
Urgency	**siku**	feel + force	Said with pressing importance
Regret	**sima**	feel + opposite	Said with sorrow for what happened
Surprise	**sipu**	feel + unknown	Said with astonishment
Reassurance	**siya**	feel + harmony	Said to comfort / encourage

Examples:
mi mu-nu siku. (I will go — urgently.)
pi ma mu-ti sima. (The person did not go — regretfully.)
tu su sa siya. (You are good — said reassuringly.)
mi tu yiwu-ti sipu. (I saw you — with surprise!)

Note: Affect particles are always optional. Omitting them produces a neutral, unmarked tone. They express the speaker's attitude toward the *utterance*, not the subject's emotion (which is expressed via modifier roots like *ya*, *wi*, *ku*).

3.11 Adjectives & Adverbs (Free Modifiers)

Roots can function as **free modifiers** (adjectives, adverbs, or emotions). To ensure absolute clarity, any root used as a modifier must be placed *immediately before* the word it describes.

- ***su*** *pi wuka.* (The good person eats.)
- *mi* ***ya*** *mu.* (I joyfully go.)
  Script: [⊖′] [⊔·] [⊖_]
- *mi* ***wi*** *na.* (I sadly speak.)
  Script: [⊖′] [⌣′] [$\overline{△}$·]
- *pi* ***ku*** *na-ti.* (The person angrily spoke.)
  Script: [◯′] [□_] [$\overline{△}$·][△′]

- **Intensity:** Prefix *wa*: *mi **wa wi** na.* (I speak with deep sorrow.)
  Script: [⊖′] [⌣ ·] [⌣′] [$\overline{\triangle}$·]
- **Mixed emotions:** Stack freely: *mi **ya wi** si.* (I feel bittersweet.)
  Script: [⊖′] [⊔·] [⌣′] [*Z*′]
- **Emotional intensifier:** *sisi* (feel-feel) frames a clause as pure, overwhelming emotion:
  mi sisi ya wi si-ti. (I felt an overwhelming bittersweetness.)
  Use *sisi* only when the speaker wishes to mark the utterance as deeply emotional, not merely descriptive.

3.12 Comparatives

- **More:** *wa* before adjective. *wa su* = better. *wa ku* = stronger.
- **Most:** *wawa* before adjective. *wawa su* = best.
- **Less:** *mawa* before adjective. *mawa su* = worse.
- **Than:** *timu* (behind-move = "beyond") after the compared noun: *mi tu timu wa su sa.* (I, you-beyond, more good am.)
  Script: [⊖′] [△_] [△′][⊖_] [⌣ ·] [*Z*_] [*Z*·]

Note: ti (past/behind) serves as a tense suffix and distance demonstrative, unified by the core metaphor of "what is behind." The comparative "than" uses the dedicated compound *timu* (beyond) to avoid overloading *ti*.

3.13 Conditionals & Modality

Conditionals: *panu* (question-forward) at the start of a clause marks it as conditional.

- *panu yumu ka, mi mu-nu.* (If rain happens, I will leave.)
  Script: [◯·][$\overline{\triangle}$_] [⊔_][⊖_] [□·] [⊖′] [⊖_][$\overline{\triangle}$_]

Modality: Compound particles placed before the main verb.

- **suka** (good-do = able): *mi suka mu.* (I can go.)
  Script: [⊖′] [*Z*_][□·] [⊖_]
- **kuka** (force-do = obliged): *mi kuka ka.* (I must do this.)
  Script: [⊖′] [□_][□·] [□·]
- **puka** (unknown-do = uncertain): *mi puka mu.* (I might go.)
  Script: [⊖′] [◯_][□·] [⊖_]

3.14 Negation

To negate an action, place the word **ma** (opposite/no) directly before the verb. This explicitly negates the clause, distinguishing it from compound antonyms (like *masupi* = enemy).

- *mi ma mu.* (I do not go.)
  Script: [⊖′] [⊖·] [⊖_]

3.15 Spatial Postpositions

Spatial relationships are marked by **postpositions** — roots placed *after* the noun they govern and before the verb, consistent with SOV order. Position in the sentence identifies their grammatical role.

English	Sunina	Root Logic
to / toward	**nu**	forward
from	**timu**	behind + move
at / in	**yi**	space
with (accompany)	**ki**	bond

Examples:
pi yusa nu mu-ti. (Person water-toward went. = The person walked to the water.)
mi nisa yi ni. (I book-at think. = I think about the book.)
mi supi ki mu. (I friend-with go. = I go with a friend.)

3.16 Imperative Mood

Commands use the particle **tuka** (target-do = "direct to act") at the start of the sentence, preserving SOV order with zero exceptions. The subject *tu* (you) is implied and may be omitted.

- *tuka mu!* (Go!)
- *tuka ma wuka!* (Don't eat!)
- *tuka tasa yiwu!* (Look at this!)

Polite request: Prefix *su* (good): *su tuka na.* (Please speak.)

Note: The imperative uses *tuka* (target + do), not *paka*, to avoid collision with the question word series (*pa-*). Each grammar particle occupies a unique form.

3.17 Reflexive & Reciprocal

- **Reflexive *kami*** (equal + self = "oneself"): placed in the object position.
  pi kami yiwu. (The person sees oneself.) *mi kami yiwu.* (I see myself.)
- **Reciprocal *kimi*** (bond + self = "each other"): placed in the object position.
  piwa kimi yiwu. (People see each other.)

3.18 Derivational Morphology

Systematic suffixes convert roots between grammatical categories. Because every suffix is itself a root with transparent meaning, speakers can apply them productively to any stem.

Conversion	Suffix	Logic	Example	Attested
Verb → Noun (act/result)	**-sa**	thing	*mu* → *musa* (journey)	30+
Verb → Agent (doer)	**-pi**	person	*na* → *napi* (speaker)	21
Adj → Noun (quality)	**-sa**	thing	*su* → *susa* (goodness)	10+
Noun/Adj → Verb (become/do)	**-mu**	change	*su* → *sumu* (improve)	5+
Noun → Action verb	**-ka**	do	*mu* → *muka* (go/walk)	30+
Root → Feeling/state	**-si**	feel	*ya* → *yasi* (happy)	14
Any → Antonym	**ma-**	opposite	*supi* → *masupi* (enemy)	30+
Any → Intensified	**wa-**	intense	*yusa* → *wayusa* (ocean)	10+

Note: Roots used directly as pre-modifiers require no suffix (see §3.11). The "Attested" column shows how many standardized words already use each pattern, confirming its productivity.

3.19 Productive Compound Templates

Beyond single-suffix derivation (§3.18), Sunina's lexicon reveals recurring **multi-root templates** — compound endings that reliably produce words in a semantic domain. These templates are now formalized as productive rules: speakers may coin new words by slotting any appropriate modifier into the **X** position without waiting for Academy approval, provided the result obeys CV phonology and passes the collision check.

Tier 1 — Freely Productive (high attestation, fully regular; coin at will)

Template	Ending	Means	Attested	Examples
X-*pi*	person	"person of/who X"	21	*supi* (friend), *nanipi* (teacher), *kamupi* (worker)
X-*ka*	do	"to do X"	30+	*muka* (go), *yisika* (look), *sakika* (build)
X-*sa*	thing	"the thing of X"	30+	*yusa* (water), *ninusa* (idea), *katusa* (success)
X-*si*	feel	"the feeling/quality of X"	14	*yasi* (happy), *wisi* (sad), *kusi* (hard)
ma-X	opposite	"anti-/un-/non- X"	30+	*masupi* (enemy), *mayumu* (drought)
wa-X	intense	"great/intense X"	10+	*wayusa* (ocean), *wawisa* (mountain)

Tier 2 — Domain Productive (regular within their semantic field; coin freely for that domain)

Template	Ending	Means	Attested	Examples
X-*muka*	move+do	"to move in X way"	14	*kumuka* (run), *yumuka* (swim), *yinumuka* (climb)
X-*kusa*	force+thing	"tool/instrument for X"	8	*wakusa* (weapon), *yikakusa* (key), *susikusa* (broom)
X-*pisa*	life+thing	"body part of X function"	18	*kapisa* (hand), *nipisa* (head), *sipisa* (heart)
X-*pimu*	life+move	"creature/animal of X"	8	*yimupimu* (bird), *yumupimu* (fish), *sukipimu* (dog)
X-*wusa*	consume+thing	"food/edible of X"	7	*yuwusa* (fruit), *wisawusa* (grain)
X-*yusa*	fluid+thing	"liquid/body-of-water of X"	10	*muyusa* (river), *wayusa* (ocean), *puyusa* (fog)
X-*kisa*	bond+thing	"structure/building of X"	6	*yikisa* (house), *yinukisa* (roof)
X-*sasa*	thing+thing	"material/substance of X"	4	*kuwisasa* (metal), *nisasa* (paper)

How to read the templates. Each template is head-final: the ending defines *what the word is* (person, tool, creature. . .), and the X-slot narrows its meaning. Multiple modifier roots may stack before the ending: *wa-ki-pi* = intense > rule > **person** = "ruler."

Coining example: To create "singer" (a person who sings), apply the **agent** template to the verb "sing" (*yanaka*): drop the verb ending and attach *-pi* → *yanapi* (sing-person). Or compose directly: *ya-na-ka-pi* (joy + speak + do + person). Both routes converge on a suggestive, head-final compound.

Coining Checklist (apply before using any newly coined word):

1. **CV compliance** — every syllable is one consonant + one vowel.
2. **Head-final** — the semantic head (template ending) is the rightmost element.
3. **Collision-free** — the full form and its short form (first 2 syllables) do not duplicate any existing word, root, particle, or approved short form. If the 2-syllable short collides, extend to 3 syllables; if that also collides, the word has no short form.
4. **Suggestive** — a listener who knows the 24 roots can infer the *semantic neighbourhood* of the compound. The decomposition should hint at the meaning, even if the precise sense requires learning the standardized definition.
5. **Minimal** — use the fewest roots that uniquely identify the concept; do not overspecify.

Words coined via Tier 1 templates are immediately usable. Words coined via Tier 2 templates are usable within their domain. Wholly novel template endings (not listed above) require Academy standardization before adoption.

3.20 Disambiguation Conventions

Several roots serve multiple grammatical roles. Sunina resolves potential ambiguity through strict writing and prosodic conventions:

1. **Suffixes are hyphenated** to their host word: *ka-ti* (made), *mu-nu* (will go). They share the host word's stress domain.

2. **Compound words are fused** into one unhyphenated word: *supi* (friend), *timu* (from/than), *kuka* (must).
3. **Free particles stand alone** as separate words with independent stress: *ki* (and), *ma* (not), *panu* (if).

Thus negation *ma ka-ti* (not + did) is visually and prosodically distinct from the passive suffix *ka-maka-ti* (bound to the verb within its stress domain). Similarly, the conjunction *ki* (standalone) is distinct from the possessive suffix *-ki* (attached to the possessor).

4 Core Vocabulary

The following words are **standardized compounds** derived from the 24 roots. They are the official forms — speakers should prefer these over ad-hoc coinages to ensure mutual intelligibility. Short forms (where listed) may be used in casual speech.

4.1 People & Relations

English	Sunina	Short	Logic
Friend	supi	—	good + person
Enemy	masupi	—	opposite + friend
Child	nupi	—	future + person
Elder	tipi	—	past + person
Baby / Infant	mawanupi	mawanu	opposite-intense + child (small child)
Parent	kipi	—	bond + person
Family	kipiwa	—	bond + people
Partner / Spouse	kisupi	kisu	bond + good + person
Sibling	kakipi	kaki	equal-bond + person
Stranger	pupi	—	unknown + person
Leader	wapi	—	intense + person
Teacher	nanipi	nani	speak-mind + person
Student	wunipi	wuni	consume-mind + person
Worker	kamupi	kamu	do-move + person
Neighbour	kiyipi	kiyi	bond-space + person
Ancestor	tikipi	tiki	past-bond + person

4.2 Nature & Environment

English	Sunina	Short	Logic
Sky	yiwa	—	space + big (expanse)
Sun	wayi	—	intense + light
Moon	mawayi	—	opposite-intense + light
Star	puyi	—	unknown + light
Cloud	yuyiwa	yuyi	fluid + sky
Wind	muyisa	—	move + air
Rain	yumu	—	fluid + move
Storm	kuyumu	kuyu	force + rain
Snow	makuyusa	maku	opposite-force + water (soft water)
Air	yisa	—	space + thing
River	muyusa	muyu	move + water
Lake	yiyusa	yiyu	space + water
Ocean	wayusa	wayu	intense + water
Mountain	wawisa	wawi	intense + earth
Stone / Rock	kuwisa	kuwi	force + earth (hard earth)
Sand	mawawisa	mawawi	opposite-intense + earth (small earth)
Mud	yuwisa	yuwi	fluid + earth
Fire	kuyi	—	force + light
Smoke	mukuyi	muku	move + fire
Ash	tikuyi	tiku	past + fire
Earth	wisa	—	heavy + thing
Tree	yupi	—	nature + life
Flower	suyupi	suyu	good + tree
Seed	nuyupi	nuyu	future + tree
Leaf	tayupi	tayu	present + tree
Thunder	nakuyu	naku	speak + storm
Lightning	yikuyu	—	light + storm
Ice / Frost	kuwayusa	kuwayu	hard + water
Fog / Mist	puyusa	puyu	unknown + water
Forest	wayupi	—	intense + tree
Field / Meadow	tawisa	—	surface + earth
Dawn	nuwayi	nuwa	forward + sun
Dusk	manuwayi	manu	opposite + dawn
Drought	mayumu	—	opposite + rain
Flood	wuyumu	—	consume + rain
Dew	miyumu	miyu	self + rain
Pond	wiyusa	wiyu	weight + water

4.3 Body & Senses

English	Sunina	Short	Logic
Body	pisa	—	life + thing
Head	nipisa	—	mind + body
Hand	kapisa	—	do + body
Foot	mupisa	—	move + body
Arm	wakapisa	wakapi	intense-do + body
Leg	wamupisa	wamu	intense-move + body
Back	tipisa	—	behind + body
Mouth	napisa	—	speak + body
Nose	yusipisa	yusi	fluid-sense + body
Tongue	nakapisa	naka	speak-do + body
Tooth	wukupisa	wuku	consume-force + body
Bone	kupisa	—	force + body (hard body)
Skin	tapisa	—	present/surface + body
Stomach	wupisa	—	consume + body
Blood	muyupisa	muyupi	move-fluid + body
Heart	sipisa	—	feel + body
Eye	yiwupisa	yiwupi	see + body
Ear	siwupisa	siwupi	hear + body

4.4 Animals

Compound animal names are **descriptive classifications**: they tell the listener *what kind of creature* is being discussed. Multi-root modifiers ensure phonological distinctiveness between species. For everyday reference to specific species, the loanword system (§4.16) is preferred: *na* + an adapted name gives each animal a short, unique, unmistakable label. The compound form remains as a generic descriptor — useful when encountering an unfamiliar creature or when a loanword has not yet been standardized.

English	Sunina	Short	Logic
Animal / Creature	pimu	—	life + move
Bird	yimupimu	yimupi	space-move + creature
Fish	yumupimu	yumupi	fluid-move + creature
Dog	sukipimu	sukipi	good-bond + creature
Cat	misupimu	misu	self-good + creature
Horse	kumupimu	kumupi	force-move + creature
Snake	wisapimu	wisapi	earth + creature
Insect	mawapimu	mawapi	opposite-intense + creature (small creature)

Note: Short forms follow the standard collision-avoidance rule: if the first two syllables collide with an existing word, the short form extends to three syllables. For instance, *yimupimu* (bird) cannot shorten to *yimu* (already "stand/rise"), so its short form is *yimupi*. Specific species beyond these core entries should be borrowed as proper-noun loanwords (*na* + adapted name) for everyday use.

4.5 Food & Drink

English	Sunina	Short	Logic
Fruit	yuwusa	—	nature + food
Grain / Bread	wisawusa	wisawu	earth + food
Meat	pimuwusa	pimuwu	animal + food
Milk	piyusa	piyu	life + water
Salt	kuwusa	kuwu	force + food
Cook	wusaka	—	food + make
Hungry	mawusa	mawu	no + food
Thirsty	mayusa	mayu	no + water
Delicious	suwusa	suwu	good + food

4.6 Core Verbs

Note: The verbs *wuka* (eat), *yuwu* (drink), and *masiyaka* (sleep) appear under §4.7 (Actions & Objects) below. Bare roots such as *mu*, *na*, *ni*, and *si* remain valid as broad predicates in grammar examples; the lexical verbs below are narrower standardized compounds used for more specific everyday meanings.

	English	Sunina	Short	Logic
Movement	Go / Walk	muka	—	move + do
	Run	kumuka	kumu	force + go
	Come	numuka	numu	forward + go
	Sit / Stay	mamuka	mamu	opposite + go
	Stand / Rise	yimuka	yimu	space + go
	Fly	yiwamuka	yiwamu	sky + go
	Swim	yumuka	—	fluid + go
	Fall	wimuka	wimu	weight + go

	English	Sunina	Short	Logic
Manipulation	Give	nuka	—	forward + do
	Take	tika	—	behind + do
	Hold / Keep	kika	—	bond + do
	Open	yika	—	space + do
	Close	mayika	mayi	opposite + open
	Push	nuku	—	forward + force
	Pull	tikuka	—	back + force + do
	Build	sakika	saki	thing + bond + do
	Break	kukika	kuki	force + bond + do

Senses & Speech

English	Sunina	Short	Logic
See	yiwu	—	light + take in
Hear	siwu	—	sense + take in
Look / Observe	yisika	yisi	light + feel + do
Listen / Attend	nasika	nasi	speak + feel + do
Say / Utter	namu	—	speak + move
Think	nika	—	mind + do
Feel	sika	—	feel + do

Life & Time

English	Sunina	Short	Logic
Live	pika	—	life + do
Die	mapika	mapi	opposite + live
Grow	mupika	—	change + live
Begin / Start	taka	—	present + do
End / Finish	mataka	mata	opposite + begin

Social & Other

English	Sunina	Short	Logic
Help	sukika	suki	good + bond + do
Fight	kuwaka	kuwa	force + intense + do
Search	nisaka	—	mind + thing + do
Teach	ninaka	nina	mind + speak + do
Learn	sinika	sini	feel + mind + do
Want / Need	simu	—	feel + move
Know	niki	—	mind + bond
Wait	tamu	—	time + move
Return	timuka	—	past + go
Carry	samuka	samu	thing + go

Activity & Physical

English	Sunina	Short	Logic
Play	yakika	—	joy + bond + do
Dance	yasika	—	joy + feel + do
Sing	yanaka	—	joy + speak + do
Wash / Bathe	yusuka	yusu	fluid + good + do
Rest	makuka	—	opposite-force + do
Dig	wisaka	—	earth + do
Plant (verb)	nuyuka	—	future + nature + do
Cut	kutika	kuti	force + back + do
Throw	nukuka	—	forward + force + do
Catch	tikika	—	back + bond + do
Climb	yinumuka	yinumu	above + go
Jump / Leap	kunumuka	kunumu	force + forward + go
Turn / Rotate	simuka	—	feel + go (shift direction)
Call / Summon	wanaka	wana	intense + speak + do
Count	nasaka	nasa	speak + thing + do

4.7 Actions & Objects

English	Sunina	Short	Logic
Eat	wuka	—	consume + do
Drink	yuwu	—	fluid + consume
Sleep	masiyaka	masi	opposite-feel + harmony + do
House	yikisa	yiki	space + rule + thing
Food	wusa	—	consume + thing
Path / Road	muyi	—	move + space
Song	yanasa	yana	joy + speak + thing

4.8 Objects & Artefacts

English	Sunina	Short	Logic
Tool	kusa	—	force + thing
Weapon	wakusa	waku	intense + tool
Knife	mawakusa	mawaku	small + tool
Rope	kimusa	kimu	bond + move + thing
Clothing	kipisa	—	bond + body
Boat	yumusa	—	fluid + move + thing
Lamp	yikusa	yiku	light + force + thing
Mirror	miyisa	miyi	self + light + thing
Gift	nukasa	—	forward-do + thing
Money	kasusa	kasu	equal + good + thing
Vehicle	kumusa	—	force + move + thing
Bed	makasa	—	opposite-do + thing
Door	yikasa	—	open + thing
Wall	sakisa	—	thing + bond + thing
Container	kikusa	kiku	bond + force + thing
Key	yikakusa	yikaku	open + tool

Household

English	Sunina	Short	Logic
Table	kasisa	kasi	equal-feel + thing (flat surface)
Chair / Seat	mamukasa	—	sit + thing
Cup / Bowl	wukisa	wuki	consume + bond + thing
Pot / Cooking vessel	wuyusa	wuyu	consume + fluid + thing
Plate / Dish	kukasa	—	force-do + thing (flat hard)
Blanket / Cover	wipisa	wipi	weight + body
Window	yisakisa	yisaki	light + structure
Roof	yinukisa	yinuki	above + structure
Floor	mayinukisa	—	below + structure
Basket	wikisa	wiki	weight + bond + thing
Pillow	nipikusa	nipiku	head + tool
Shelf	sasisa	sasi	thing-feel + thing (display surface)
Broom / Brush	susikusa	susiku	clean + tool
Candle	yikuyisa	yikuyi	fire-light + thing

4.9 Time & Place

English	Sunina	Short	Logic
Day	yita	—	light + time
Night	mayita	—	opposite + day
Morning	tanu	—	time + forward
Evening	tati	—	time + past
Tomorrow	nuyita	nuyi	forward + day
Yesterday	tiyita	—	past + day
Year	yutasa	yuta	nature + time + thing
Always	wata	—	intense + time
Never	mawata	—	opposite + always
City / Town	wayiki	—	intense + space + rule
Now	tasi	—	present + sense
Soon	nutamu	nuta	forward + wait
Later	puta	—	unknown + present
Before (temporal)	kita	—	bond + time
After (temporal)	nuki	—	forward + bond
Week	manayita	manayi	seven + day
Month	mawayita	—	moon + time
Season	yuwata	yuwa	nature + intense + time
Moment / Instant	sitamu	sita	sense + wait

4.10 Spatial & Direction

These compounds function as spatial postpositions (§3.15) — placed *after* the noun they govern and before the verb.

English	Sunina	Short	Logic
Above / Up	yinu	—	space + forward
Below / Down	mayinu	—	opposite + above
Inside / Within	niyi	—	inner + space
Outside	maniyi	mani	opposite + inside
Near / Close to	siyi	—	sense + space
Far from	masiyi	—	opposite + near
Between / Among	kayi	—	equal + space
Behind (spatial)	yiti	—	space + behind

4.11 Quantity & Determiners

Quantifiers and determiners occupy the same pre-noun modifier position as adjectives (§3.11). They combine naturally with plural *-wa* and demonstratives.

English	Sunina	Short	Logic
All / Every	kiwa	—	bond + intense (totally connected)
Some	pusa	—	unknown + thing
None	makiwa	—	opposite + all
Many / Much	puwa	—	unknown + intense (unknowably large)
Few	mapuwa	mapu	opposite + many
Enough / Sufficient	suwa	—	good + intense
Part / Piece	makisa	—	opposite-bond + thing
Whole / Complete	kawasa	kawa	equal + intense + thing
Only / Sole	sakami	saka	thing + equal + self
Other / Another	musi	—	change + sense
Same	kaka	—	equal + equal
Different	makaka	—	opposite + same

Examples:
kiwa piwa mu-ti. (All people went.)
mi pusa nisa yiwu-ti. (I some book saw. = I saw some book.)
pi musi sa yiwu-nu. (The person other thing see-will. = The person will see another thing.)

4.12 Qualities & States

English	Sunina	Short	Logic
Bad	masu	—	opposite + good
New	nusa	—	future + thing
Old (age)	watisa	wati	intense-past + thing
Hot	sikuyi	—	feel + fire
Cold	masikuyi	masiku	opposite + hot
True	sunisa	suni	good + mind + thing
False	masunisa	masuni	opposite + true
Beautiful	suyisa	suyi	good + light + thing
Full	wasa	—	intense + thing
Empty	mawasa	—	opposite + full
Strong	kunu	—	force + forward
Weak	makunu	—	opposite + strong
Long / Tall	wamuyi	—	intense + move + space
Dark	mayisa	—	opposite + light + thing
Wet	yuku	—	fluid + force
Dry	mayuku	—	opposite + wet
Clean / Pure	susi	—	good + feel
Dirty	masusi	—	opposite + clean
Hard (texture)	kusi	—	force + feel
Soft	makusi	—	opposite + hard
Heavy	wiku	—	weight + force
Light (weight)	mawiku	mawi	opposite + heavy
Fast / Quick	muwa	—	move + intense
Slow	mamuwa	—	opposite + fast
Loud	nawi	—	speak + weight
Quiet / Silent	manawi	—	opposite + loud
Deep	puwi	—	unknown + weight
Shallow	mapuwi	—	opposite + deep
Wide / Broad	tawi	—	present + weight (spread out)
Narrow	matawi	—	opposite + wide

4.13 Emotions & Discourse

English	Sunina	Short	Logic
Happy	yasi	—	joy + feel
Sad	wisi	—	sorrow + feel
Angry	wasi	—	intense + feel
Afraid	pusi	—	unknown + feel
Love	suyasi	suya	good + happy
Hate	masuyasi	masuya	opposite + love
Hope	nuyasi	nuya	forward + happy
Peace / Harmony	yaki	—	joy + bond
Pride	wanisi	wani	intense + mind + feel
Thank	sunaka	—	good + speak + do
Sorry / Apology	wisika	—	sorrow + feel + do
Farewell	yamuka	yamu	joy + go

4.14 Abstract & Mental Concepts

English	Sunina	Short	Logic
Idea	ninusa	ninu	mind + forward + thing
Reason / Cause	kitisa	—	bond-past + thing
Problem	kunisa	kuni	force + mind + thing
Answer / Reply	panasa	—	question + speak + thing
Solution	sukunisa	suku	good + problem
Power	kukisa	—	force + bond + thing
Freedom	makinu	—	opposite + bond + forward
Wisdom	wanisa	—	intense + mind + thing
Skill / Ability	sukasa	—	good + do + thing
Chance / Luck	pukisa	puki	unknown + bond + thing
Purpose / Goal	tuni	—	target + mind
Mistake / Error	manika	—	opposite + mind + do
Success	katusa	katu	equal + target + thing
Failure	makatusa	—	opposite + success
Method / Way	nimu	—	mind + move

4.15 Society, Colours & Health

	English	Sunina	Short	Logic
Society	King / Ruler	wakipi	waki	intense + rule + person
	Law	kisa	—	rule + thing
	War / Conflict	kuwasa	—	force + intense + thing
	Trade / Exchange	kasamu	kasa	equal + thing + move
	Story / History	tinisa	tini	past + mind + thing
	Dream	niyasi	niya	mind + joy + feel
	Danger	masusa	—	bad + thing

Materials

English	Sunina	Short	Logic
Wood	yupisa	—	nature + body + thing
Metal	kuwisasa	—	force + earth + substance
Cloth / Fabric	kisasa	—	bond + substance
Paper	nisasa	—	mind + substance
Glass	yisisa	—	light + feel + thing
Clay	kawisa	kawi	do + earth + thing
Leather	pimusasa	pimusa	creature + substance

Colours

English	Sunina	Short	Logic
Red	kuyisa	—	fire + thing
Blue	yiwasa	—	sky + thing
Green	yuyisa	—	nature + light + thing
White	yisusa	yisu	light + good + thing
Black	mayisusa	mayisu	opposite + white
Yellow	wayisa	—	intense + light + thing
Brown	wisayisa	wisayi	earth + light + thing
Grey	kayisa	—	balanced + light + thing
Orange	wakuyisa	wakuyi	intense + fire + thing

Health

English	Sunina	Short	Logic
Healthy	supisa	—	good + body
Sick / Ill	masupisa	—	opposite + healthy
Heal	supika	—	good + life + do
Medicine	supikasa	—	heal + thing

4.16 Proper Nouns & Loanwords

Foreign names and borrowed terms are adapted to Sunina's CV phonology:

1. Map each foreign sound to its nearest Sunina consonant and vowel.
2. Break consonant clusters by inserting the default vowel *a*.
3. Drop word-final consonants or append the default vowel *a*.

The particle **na** (word/name) precedes a proper noun to mark it as a name: *na Pakisi* = Paris. *na Yuna* = John.

Lexicon Note: The standardized vocabulary contains over 300 entries across fifteen semantic subsections. New words should follow the same derivation principles and be standardized by community consensus before widespread use. An official Sunina Academy and dictionary govern formal compound definitions and authorized short forms; mutual intelligibility depends on speakers sharing the same compounds rather than coining ad-hoc alternatives.

5 Base-10 Mathematics

All base digits and place-value multipliers end in **-na**, making cardinal numbers systematic and easy to parse.

5.1 The Number Grid (0–9)

The vowel cycle $a \to i \to u$ repeats across three consonant blocks.

S-Block (Low)	K-Block (Mid)	M-Block (High)
1 sana	4 kana	7 mana
2 sina	5 kina	8 mina
3 suna	6 kuna	9 muna

Zero (0): **nuna** (*nu* [forward] + *-na* [number]: the forward-empty state — nothing yet exists).

Logic: If you know K-block = mid and u = third vowel, then *kuna* must be 6.

5.2 Place Values & Operators

Multiplier	Word	Operator	Word
×10	pana	+ (Plus)	su
×100	pina	− (Minus)	masu
×1,000	puna	× (Multiply)	waka
		÷ (Divide)	mawaka
		= (Equals)	ka

Examples:
305: suna pina kina. (3 × 100, 5. Digit–multiplier pairs are multiplied; consecutive groups sum by juxtaposition. Zero places are omitted.)
Script: [Z_][$\overline{\triangle}$·] [◯′][$\overline{\triangle}$·] [□′][$\overline{\triangle}$·]

1 + 2 = 3: sana su sina ka suna.
Script: [Z·][$\overline{\triangle}$·] [Z_] [Z′][$\overline{\triangle}$·] [□·] [Z_][$\overline{\triangle}$·]

Note: Operators such as *su*, *masu*, and *ka* appear strictly between number-words in mathematical expressions and are unambiguous in context. The **P** root (Life/Unknown) is used for powers of ten (*pana, pina, puna*), serving as neutral positional multipliers in base-10 math.

6 The Featural Writing System

Letter shapes mimic the shape of the mouth producing the sound. The consonant dictates the base geometric shape; the vowel acts as a modifier mark.

6.1 Base Shapes (Consonants)

- **p** (Lips): ◯ **m** (Lips/Hum): ⊖ **w** (Lip Glide): ◡

- **t** (Tongue Tip): $\triangle$ **n** (Tongue/Hum): $\overline{\triangle}$ **s** (Hiss): Z
- **k** (Throat): $\square$ **y** (Open Throat): $\sqcup$

6.2 Vowel Modifiers

Each vowel occupies a fixed position within the consonant's glyph block:

- **-a**: Centered dot ($\cdot$) **-i**: Top vertical tick ($'$) **-u**: Bottom horizontal bar ($_$)

Each CV syllable occupies one fixed-size square block, making the script monospaced and scannable — similar in principle to Korean Hangul.

Handwriting Guidance: In rapid handwriting, the three vowel marks may be exaggerated for clarity: the centered dot ($\cdot$) may be drawn as a small solid circle; the top tick ($'$) as a short upward stroke extending above the glyph box; and the bottom bar ($_$) as a small downward curve beneath it. This allographic variation preserves legibility at speed without changing the script's logic.

Font Development: The placeholder glyphs used throughout this document (e.g., [◯′]) are schematic representations. A dedicated Sunina font mapping each CV syllable to a proper vector glyph is planned for a future appendix.

6.3 Complete Glyph Table

	-a ($\cdot$)	**-i** ($'$)	**-u** ($_$)
p ◯	[◯·]	[◯′]	[◯_]
t △	[△·]	[△′]	[△_]
k □	[□·]	[□′]	[□_]
m ⊖	[⊖·]	[⊖′]	[⊖_]
n $\overline{\triangle}$	[$\overline{\triangle}$·]	[$\overline{\triangle}'$]	[$\overline{\triangle}$_]
s Z	[Z·]	[Z']	[Z_]
w ◡	[◡ ·]	[◡′]	[◡ _]
y ⊔	[⊔·]	[⊔′]	[⊔_]

6.4 Orientation

Standard: Horizontal left-to-right, matching the majority of the world's writing systems and all digital interfaces.
Artistic/Ceremonial: Vertical top-to-bottom (columns left-to-right), reserved for calligraphy.

6.5 Punctuation

Bracket Convention: Square brackets [ ] always denote a **syllable block** (a spoken sound). Round parentheses ( ) always denote a **punctuation mark** (a silent symbol). This distinction ensures that syllables and punctuation are never visually confused.

- **Word boundary:** Space (gap between glyph blocks of separate words; syllables within a single word are written contiguously).

- **Sentence end:** | (vertical bar).
- **Question:** (◯·) at end of sentence — the *pa* glyph in round parentheses.
- **Quotation:** ‖ before and after quoted speech.

7 Story Translation: “The Meeting”

English:
The person walked to the water. The person saw a fierce creature. The creature did not attack, because the creature was a friend. The person felt deep joy.

Sunina (Romanized):
pi yusa nu mu-ti. |
pi wakupimu yiwu-ti. |
wakupimu ma kuwaka-ti, kiti wakupimu supi ka-ti. |
pi wa ya si-ti. |

Horizontal Script (contiguous blocks per word, separated by word spaces):
S1: [◯′] [⊔_][Z·] [$\overline{\triangle}$_] [⊖_][△′] |
S2: [◯′] [⌣ ·][□_][◯′][⊖_] [⊔′][⌣ _][△′] |
S3: [⌣ ·][□_][◯′][⊖_] [⊖·] [□_][⌣ ·][□·][△′] [□′][△′] [⌣ ·][□_][◯′][⊖_] [Z_][◯′] [□·][△′] |
S4: [◯′] [⌣ ·] [⊔·] [Z′][△′] |